

PAPER Out-of-Distribution Detection and Epistemic Gating for Autonomous Fusion Control: A Machine-Readable Extrapolation Ledger That Throttles Model Authority Beyond the Validated Envelope

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Abstract

A learned controller extrapolates smoothly and confidently into regions where it has no evidence, and in a nuclear system that confident wrongness off-distribution is the dominant failure mode. The safeguard is not a better model but an honest one: a controller that measures the boundary of its own competence in real time and refuses authority beyond it. We formalise the epistemic-gating layer of the Kronos autonomous fusion controller and reduce it to a set of computable inequalities. The plant state is scored against a *mapped* validated envelope — a nine-point design-closure grid spanning fusion gain $Q = 2.735\text{--}3.460$ and fusion power $P_{\text{fus}} = 58.6\text{--}118.9\text{ MW}$ with all physics gates satisfied and the frozen design point ($Q = 3.076$, $P_{\text{fus}} = 85.04\text{ MW}$) interior — by two complementary detectors: a reconstruction/feature-space out-of-distribution (OOD) score and a machine-readable *extrapolation ledger* resolved across a nine-device breeder-and-burner post-diction base. Authority is proportioned to *calibrated* epistemic uncertainty, decomposed from aleatoric noise through a deep-ensemble total-variance law and wrapped in a distribution-free conformal coverage guarantee; the guarded confinement model carries a posterior $H_{98} = 1.007 \pm 0.030$ from a 200-evaluation surrogate-accelerated Bayesian-optimization campaign, and the verification suite discharges 19/19 formal anchors. A single composite rule — the authority ceiling is the minimum of the OOD, ledger and uncertainty gates — throttles the learned controller smoothly toward a deterministic control-barrier floor as the operating point approaches, and then leaves, the envelope. We state the governing detector equations, the numerical formulation (autoencoder threshold calibration, Gaussian-process Bayesian optimization, envelope corner mapping, formal verification), and the GPU-accelerated campaign that produced every quantity, and we benchmark the gating stack against the out-of-distribution-detection, uncertainty-calibration and disruption-prediction literature. The result is the credibility layer beneath the autonomy: authority bounded by evidence, degrading gracefully rather than silently. The one honest gate — tightening the reconstruction threshold against machine diagnostics rather than the twin’s validated distribution — is deferred to first operating hours, and the gate is conservative by construction in the interim.

Keywords: out-of-distribution detection; extrapolation ledger; epistemic uncertainty; authority gating; conformal prediction; deep ensembles; conservative control; validated envelope; spherical tokamak

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1 Introduction

1.1 Autonomous control of a compact spherical tokamak

The Kronos breeder Hyperion is a compact, high-power-density negative-triangularity spherical tokamak: design point plasma current $I_p = 9.66$ MA, fusion gain $Q = 3.076$, fusion power $P_{\text{fus}} = 85.04$ MW, triangularity $\delta = -0.30$, on-axis field $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, and a resistive-copper centrepole that peaks at 16.84 T. The companion D-³He tandem-mirror burner (Aegis / MetroVolt) operates at engineering gain $Q_E = 1.318$ with a neutron power fraction $f_n = 5.44\%$, a 26.49 T high-field superconducting plug, and a direct-energy converter of efficiency $\eta_{\text{DEC}} = 0.49$; the breeder centrepole and the burner plug are two *distinct* magnets, and the ledger of §2.3 keeps them separate. A machine at this density leaves little time to react and no tolerance for an unmodelled excursion, so its control stack is model-based and increasingly autonomous, from real-time equilibrium reconstruction and profile prediction to learned magnetic control [1, 2, 3, 4]. The four coupled Kronos digital twins run ahead of the plant as a predictive shadow [32], and neural-operator flux and equilibrium surrogates supply the speed that closed-loop control demands [33, 22, 23].

1.2 The off-distribution failure mode

The characteristic and most dangerous failure of a learned controller is not high variance but confident error where it has no data: a smooth interpolant extrapolates past its evidence with undiminished — often *increasing* — apparent confidence [14, 26, 13]. In a nuclear plant this is inadmissible. The remedy is a system that (i) recognises, per control tick, when the live state is unlike anything the models were validated on; (ii) records, in a machine-readable form, exactly where each model has evidence; and (iii) hands authority back to a deterministic floor *before* a model is trusted out of its depth. This is the classical separation of *epistemic* uncertainty — reducible ignorance, “the model does not know” — from *aleatoric* noise — irreducible scatter [12, 17]: the safety layer must act on the former and tolerate the latter.

1.3 Prior art

Out-of-distribution detection is a mature subfield. Baseline softmax-confidence detectors [8], feature-space Mahalanobis detectors [9], deep-ensemble predictive uncertainty [10], Monte-Carlo-dropout Bayesian approximations [11] and reconstruction-based anomaly detectors [16, 15] each supply a score that separates in-distribution from novel inputs. Distribution-free conformal prediction turns any such score into a finite-sample coverage guarantee [18, 19]. In fusion, data-driven disruption predictors on DIII-D, C-Mod and EAST demonstrate that a learned warning can be both accurate and cross-machine [5, 6, 7], and control-theoretic safety filters — control-barrier functions rendering a safe set forward-invariant [24] — provide the deterministic fallback. What has been missing is the *integration*: a single, auditable rule that fuses an OOD detector, a machine-readable record of model evidence, and a calibrated uncertainty into a runtime authority ceiling for a fusion controller, grounded in a frozen, independently reproduced design register.

1.4 Contribution

This paper supplies that integration for the Kronos controller and states it as mathematics. We: (1) define the validated envelope operationally from the Kronos design-closure corner mapping (anchor KX-L1-A1) and give the formal in-distribution predicate (§2.1); (2) construct two complementary detectors — a reconstruction/feature OOD score (serial KFE-CF-SH-040) and an extrapolation-ledger support measure resolved over nine devices spanning the breeder and burner (anchor KX-L1-A7; monitor serial KFE-CF-SH-075) — and derive their calibration (§§2.2–2.3); (3) decompose predictive uncertainty into epistemic and aleatoric parts, attach a distribution-free conformal guarantee, and read the guarded confinement posterior $H_{98} = 1.007 \pm 0.030$ directly from the surrogate-accelerated Bayesian-optimization campaign (anchor KX-L1-A14, §2.4); (4) fuse the three into a single composite authority-throttle law with a deterministic control-barrier floor (§2.5); and (5) verify the stack against the formal V&V suite (19/19 anchors, KX-L1-A6) and against the published OOD, calibration and disruption-prediction benchmarks (§§4–5). Every quantitative claim traces to a frozen, reproduced anchor in the Kronos physics de-risking register [27], and the whole campaign ran on the NVIDIA GPU HPC campaign described in §3. This paper is the epistemic-gating member of the Kronos autonomy series, sitting between the calibrated-uncertainty-and-abstention result [28] and the runtime-assurance-with-signed-ledger result [29], and beneath the maturity-gated actuation-authority argument [30].

2 Governing equations and the formal gating problem

2.1 Plant, controller and the validated envelope

Write the plant as a discrete-time stochastic system on state $\mathbf{x} \in \mathbb{R}^{n_x}$ with actuation $\mathbf{u} \in \mathbb{R}^{n_u}$,

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_w), \quad (1)$$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_v), \quad (2)$$

with $\mathbf{y} \in \mathbb{R}^{n_y}$ the diagnostic vector. The controller proposes an action $\mathbf{u}_k^{\text{AI}} = \pi_\theta(\mathbf{y}_{0:k})$ from a learned policy π_θ whose transition and observation models are validated only on a bounded region of operating space. That region is not asserted; it is *mapped*. The Kronos design-closure study (KX-L1-A1) evaluates the coupled 0-D/1-D closure over a nine-point corner grid $\{\xi_j\}_{j=1}^9$ and records, for each node, whether all physics gates (power balance, stability, heat-flux, breeding and current-drive) are satisfied. The gates hold across the grid, whose projection onto the observable pair (Q, P_{fus}) spans

$$Q \in [2.735, 3.460], \quad P_{\text{fus}} \in [58.6, 118.9] \text{ MW}, \quad (3)$$

with the frozen design point $\xi^* = (Q^*, P_{\text{fus}}^*) = (3.076, 85.04 \text{ MW})$ in the interior (figure 1). Define the validated envelope $\mathcal{E}$ as the closed convex hull of the mapped grid in a whitened design coordinate, and the in-distribution predicate

$$\text{ID}(\mathbf{x}) \equiv [\Phi(\mathbf{x}) \in \mathcal{E}], \quad \Phi : \mathbb{R}^{n_x} \rightarrow \mathbb{R}^d \quad (4)$$

where Φ is the projection of the state onto the design-closure coordinates. Operation with $\text{ID}(\mathbf{x}) = \text{true}$ is supported by design closure; operation with $\text{ID}(\mathbf{x}) = \text{false}$ is, by definition, extrapolation. The gating problem is to bound the authority granted to π_θ by a continuous, calibrated proxy for how far outside $\mathcal{E}$ the live point has moved.

2.2 The out-of-distribution score

We use a whitening map that renders $\mathcal{E}$ isotropic. Let $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ be the mean and covariance of the validated design points in Φ -space; the affine whitening $\mathbf{T}(\mathbf{x}) = \boldsymbol{\Sigma}^{-1/2}(\Phi(\mathbf{x}) - \boldsymbol{\mu})$ gives the feature-space (Mahalanobis) score [9]

$$s_M(\mathbf{x}) = \|\mathbf{T}(\mathbf{x})\|_2 = \sqrt{(\Phi(\mathbf{x}) - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\Phi(\mathbf{x}) - \boldsymbol{\mu})}, \quad (5)$$

and, normalising each closure coordinate to its mapped half-range Δ_i , the box-referenced score

$$s(\mathbf{x}) = \|\mathbf{T}(\mathbf{x})\|_\infty = \max_i \frac{|\Phi_i(\mathbf{x}) - \mu_i|}{\Delta_i}, \quad (6)$$

so that $s < 1$ is interior, $s = 1$ is the envelope boundary and $s > 1$ is out-of-distribution (figure 2). Equation (6) is the geometric detector; the learned detector (SH-040) complements it. A convolutional autoencoder $g_\phi \circ e_\phi$ is trained on the validated state distribution to minimise reconstruction error, and flags a live state by the residual

$$\varepsilon_r(\mathbf{x}) = \|\mathbf{x} - g_\phi(e_\phi(\mathbf{x}))\|_2. \quad (7)$$

A state far from the training manifold reconstructs poorly, so ε_r is large exactly where the models lose evidence [16, 15]. The detector *threshold* τ is calibrated, not chosen, by fixing a target false-positive rate α_{fp} on held-out in-distribution data,

$$\tau = \inf \left\{ t : \widehat{\text{Pr}}[\varepsilon_r(\mathbf{x}) > t \mid \text{ID}(\mathbf{x})] \leq \alpha_{\text{fp}} \right\} = \widehat{q}_{1-\alpha_{\text{fp}}}(\varepsilon_r), \quad (8)$$

the empirical $(1 - \alpha_{\text{fp}})$ quantile of the in-distribution reconstruction error (figure 3). The trip predicate is $\varepsilon_r(\mathbf{x}) > \tau$.

Graceful degradation. Diagnostics drop out under radiation and off-normal events, so the detector must hold its judgement with missing channels. Partition the observation index set into observed O and missing M ; the detector conditions on the observed sub-vector by reconstructing over O only,

$$\varepsilon_r^O(\mathbf{x}) = \|\mathbf{x}_O - [g_\phi \circ e_\phi]_O(\mathbf{x}_O)\|_2, \quad \tau_O = \widehat{q}_{1-\alpha_{\text{fp}}}(\varepsilon_r^O), \quad (9)$$

with a channel-count-indexed threshold family $\{\tau_O\}$ so that the false-positive rate is held constant as $|M|$ grows (figure 4). The acceptance requirement is that the detector flag a held-out regime it was never trained on, and, on that flag, drop the controller to conservative control.

2.3 The extrapolation ledger

The OOD score is the fast tripwire; the extrapolation ledger is the slower, authoritative record of *where* each model has evidence. The Kronos multi-device post-diction (KX-L1-A7) resolves a nine-device ledger, spanning the breeder and the burner, that records for each device d and validated case a support kernel in design space. The ledger support at a query point is the kernel-weighted case density

$$\rho(\mathbf{x}) = \frac{1}{Z} \sum_{d=1}^9 \omega_d \sum_{c \in \mathcal{C}_d} K_\ell(\Phi(\mathbf{x}) - \Phi(\mathbf{x}_{d,c})), \quad K_\ell(\mathbf{r}) = \exp\left(-\frac{1}{2} \mathbf{r}^\top \ell^{-2} \mathbf{r}\right), \quad (10)$$

with device weights ω_d , per-device validated case sets $\mathcal{C}_d$, bandwidth ℓ and normaliser Z . A point with dense support (ρ high) is granted full model authority; a point at the edge (intermediate ρ) is granted reduced authority with widened uncertainty; a point with no support ($\rho \rightarrow 0$) is denied model authority and handed to the deterministic floor. The ledger is queried at run time and is machine-readable: it returns, with each verdict, the devices and cases that support (or fail to support) a prediction, so the authority decision is auditable case by case (figure 8) and closes under the runtime-assurance provenance record [29].

2.4 Epistemic uncertainty, calibration and conformal coverage

Authority is proportioned to *calibrated* uncertainty, not to raw model confidence, and only the epistemic part should throttle a model. For a deep ensemble of E members $\{p_e(y|\mathbf{x})\}_{e=1}^E$ with member means $\hat{\mu}_e$ and variances $\hat{\sigma}_e^2$ [10], the law of total variance splits the predictive variance into an aleatoric mean-of-variances and an epistemic variance-of-means,

$$\underbrace{\text{Var}[y|\mathbf{x}]}_{\text{predictive}} = \underbrace{\frac{1}{E} \sum_{e=1}^E \hat{\sigma}_e^2(\mathbf{x})}_{\text{aleatoric}} + \underbrace{\frac{1}{E} \sum_{e=1}^E (\hat{\mu}_e(\mathbf{x}) - \bar{\mu}(\mathbf{x}))^2}_{\text{epistemic } \sigma_{\text{ep}}^2(\mathbf{x})}, \quad \bar{\mu} = \frac{1}{E} \sum_e \hat{\mu}_e. \quad (11)$$

The epistemic term σ_{ep}^2 grows off-distribution — ensemble members disagree where they were never constrained — and it is σ_{ep} , not the aleatoric floor, that the gate reads. Calibration is enforced by the coverage condition: the empirical coverage of the model's α -credible intervals must match nominal, against Monte-Carlo truth,

$$|\text{Cov}_{\text{emp}}(\alpha) - \alpha| \leq 3\% \quad \forall \alpha, \quad (12)$$

and upgraded to a distribution-free *guarantee* by conformal prediction: with exchangeable calibration residuals and nonconformity score $R(\mathbf{x}, y)$, the prediction set $\hat{C}(\mathbf{x}) = \{y : R(\mathbf{x}, y) \leq \hat{q}_{1-\alpha}\}$ built from the $\lceil (1-\alpha)(N_{\text{cal}}+1) \rceil / N_{\text{cal}}$ empirical quantile of calibration scores satisfies [18, 19]

$$\Pr[y \in \hat{C}(\mathbf{x})] \geq 1 - \alpha, \quad (13)$$

with no distributional assumption and valid in finite samples. The guarded confinement model is calibrated by the surrogate-accelerated Bayesian-optimization campaign (KX-L1-A14): a 200-evaluation study yields a tight, well-centred confinement posterior

$$H_{98} = 1.007 \pm 0.030 \quad (14)$$

(figure 5), a 95% credible interval of [0.948, 1.066] that the gate reads directly. The same campaign illustrates precisely the hazard the gate exists for: its unconstrained optimizer, allowed to widen the priors and push H_{98} to a bound of 1.1, reports a bound-limited best-case gain of $Q \approx 8.3$ at $P_{\text{fus}} \approx 61$ MW — an aggressive extrapolation far outside the mapped envelope of equation (3). The gate refuses that excursion authority and holds the controller at the anchored, calibrated point.

2.5 The composite authority-throttle law and the deterministic floor

Each detector defines an authority ceiling in $[0, 1]$. From the OOD score (6), with throttle onset s_0 ,

$$A_{\text{ood}}(\mathbf{x}) = \begin{cases} 1, & s(\mathbf{x}) \leq s_0, \\ \frac{1-s(\mathbf{x})}{1-s_0}, & s_0 < s(\mathbf{x}) < 1, \\ 0, & s(\mathbf{x}) \geq 1; \end{cases} \quad (15)$$

from the ledger support (10) and the epistemic width (11), the saturating ceilings

$$A_{\text{led}}(\mathbf{x}) = \sigma_L(\rho(\mathbf{x}) - \rho_0), \quad A_\sigma(\mathbf{x}) = \left[1 - \frac{\sigma_{\text{ep}}(\mathbf{x})}{\sigma_{\text{max}}} \right]_+, \quad (16)$$

with σ_L a logistic and $[\cdot]_+$ the positive part. The composite authority is the *minimum* — the most conservative gate wins —

$$A(\mathbf{x}) = \min \{ A_{\text{ood}}(\mathbf{x}), A_{\text{led}}(\mathbf{x}), A_\sigma(\mathbf{x}) \} \in [0, 1], \quad (17)$$

and the command actually applied blends the learned action with the deterministic floor in proportion to authority,

$$\mathbf{u}_k = A(\mathbf{x}_k) \mathbf{u}_k^{\text{AI}} + (1 - A(\mathbf{x}_k)) \mathbf{u}_k^{\text{floor}}, \quad (18)$$

so that $A = 1$ is full learned authority and $A = 0$ is the pure floor (figure 6). The floor is not a constant but a certified conservative controller whose action is the control-barrier projection of a safe reference: given a barrier h with safe set $\mathcal{C}_{\text{safe}} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}$,

$$\mathbf{u}^{\text{floor}} = \arg \min_{\mathbf{u}} \|\mathbf{u} - \mathbf{u}^{\text{ref}}\|_2^2 \quad \text{s.t.} \quad L_f h(\mathbf{x}) + L_g h(\mathbf{x}) \mathbf{u} \geq -\gamma(h(\mathbf{x})), \quad (19)$$

which renders $\mathcal{C}_{\text{safe}}$ forward-invariant [24]: the plant cannot leave the safe set even when the learned controller is wrong. The certified safety filter is developed in the companion result [31]; here it is the terminal authority of equation (18). Equations (6)–(19) are the complete gating law; §4 reports the reproduced evidence behind each layer.

3 Computational methodology: the NVIDIA GPU HPC campaign

Every quantity in §2 was produced by the Kronos NVIDIA GPU HPC campaign, run on an A100/H100/A10-class accelerator fleet through the Kronos Research Toolkit; the fast protective loops (the geometric OOD score, the ledger lookup, the self-consistency monitor and the orchestrator) run on the in-house HPC digital-twin node so that they never contend with the training fleet. No compute cost enters this paper.

3.1 Envelope corner mapping (KX-L1-A1)

The validated envelope (3) is the projection of a design-closure corner map. The coupled 0-D power balance and 1-D transport closure is evaluated at the nine grid nodes $\{\xi_j\}$; at each node the physics gates are checked and the pair (Q, P_{fus}) recorded. The closure is solved by fixed-point iteration on the operating point to a relative residual $\leq 10^{-6}$; three of the nine corners run over the temperature-duty limit and are retained as boundary markers, fixing the mapped extremes of equation (3). The corner map defines $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$ and the half-ranges Δ_i used in the scores (5)–(6).

3.2 Reconstruction-detector training and threshold calibration

The autoencoder $g_\phi \circ e_\phi$ of equation (7) is trained by stochastic gradient descent (Adam, decoupled weight decay) on the validated state distribution generated by the twin, minimising the mean squared reconstruction error with an early-stopping criterion on a held-out split; training is GPU-accelerated on the fleet. The threshold τ and its channel-indexed family $\{\tau_O\}$ are then set by the empirical-quantile rule of equations (8)–(9) at the target false-positive rate, so calibration is a deterministic post-processing step on held-out reconstruction errors rather than a trained parameter. Detector inference is a single forward pass evaluated within one control tick.

3.3 Surrogate-accelerated Bayesian optimization and UQ (KX-L1-A14)

The confinement posterior (14) comes from a Gaussian-process Bayesian-optimization campaign [20, 21]. A GP with an anisotropic squared-exponential kernel is fit to a 200-evaluation design: an initial space-filling set that includes the frozen anchor, followed by expected-improvement-guided evaluations of a surrogate confinement/gain model. The GP posterior at a query $\boldsymbol{\xi}$ is Gaussian,

$$\mu_{\text{GP}}(\boldsymbol{\xi}) = \mathbf{k}(\boldsymbol{\xi})^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, \quad (20)$$

$$\sigma_{\text{GP}}^2(\boldsymbol{\xi}) = k(\boldsymbol{\xi}, \boldsymbol{\xi}) - \mathbf{k}(\boldsymbol{\xi})^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}(\boldsymbol{\xi}), \quad (21)$$

and the expected-improvement acquisition over the incumbent y^* is

$$\text{EI}(\boldsymbol{\xi}) = (y^* - \mu_{\text{GP}}) \Phi_{\mathcal{N}}\left(\frac{y^* - \mu_{\text{GP}}}{\sigma_{\text{GP}}}\right) + \sigma_{\text{GP}} \varphi_{\mathcal{N}}\left(\frac{y^* - \mu_{\text{GP}}}{\sigma_{\text{GP}}}\right), \quad (22)$$

with $\Phi_{\mathcal{N}}, \varphi_{\mathcal{N}}$ the standard normal CDF and PDF. Marginalising the GP posterior over the validated design region yields the confinement posterior of equation (14); the coverage of its credible intervals is checked against Monte-Carlo truth to the 3% tolerance of equation (12). The same run records the bound-limited optimizer excursion ($Q \approx 8.3$) cited in §2.4.

3.4 Formal verification and validation (KX-L1-A6)

The suite that gates model promotion is the formal verification-and-validation battery: the method of manufactured solutions (MMS) for code order-of-accuracy, the grid-convergence index (GCI) for discretization error [25], variance-based Sobol sensitivity for input attribution, and a predictive-capability maturity-model (PCMM) score for readiness. The suite discharges 19/19 anchors PASS (figure 7); MMS recovers the design order to within tolerance, GCI falls below the acceptance bound under refinement, the Sobol indices converge, and the PCMM entries clear their thresholds. These are the discharged proofs the ledger monitor (SH-075) requires before it certifies a lineage.

3.5 Reproducibility and scale

The four gating anchors are frozen entries in the register [27] and regenerable from the archived closure, UQ and verification runs. The problem scale is set by the training and ensemble-forecast workloads on the GPU fleet and the quantile-calibration and ledger-kernel evaluations on the twin node; each figure in §4 is produced by a single deposited generation script from the frozen anchor values. The independent-library reproduction discipline is the subject of the verification-backbone companion [35].

4 Results

4.1 The validated envelope and the gate score field

Figure 1 shows the mapped envelope in the (Q, P_{fus}) plane: the nine corner-mapping evaluations span $Q = 2.735\text{--}3.460$ and $P_{\text{fus}} = 58.6\text{--}118.9$ MW, and the frozen design point (3.076, 85.04 MW) sits in the interior. Query points outside the mapped box are flagged out-of-distribution and denied model authority. Figure 2 renders the continuous gate score $s(\mathbf{x})$ of equation (6) over the same plane: the interior is low-score, the boundary $s = 1$ is the hard gate, and the throttle onset $s = s_0$ is the inner contour where authority begins to fall.

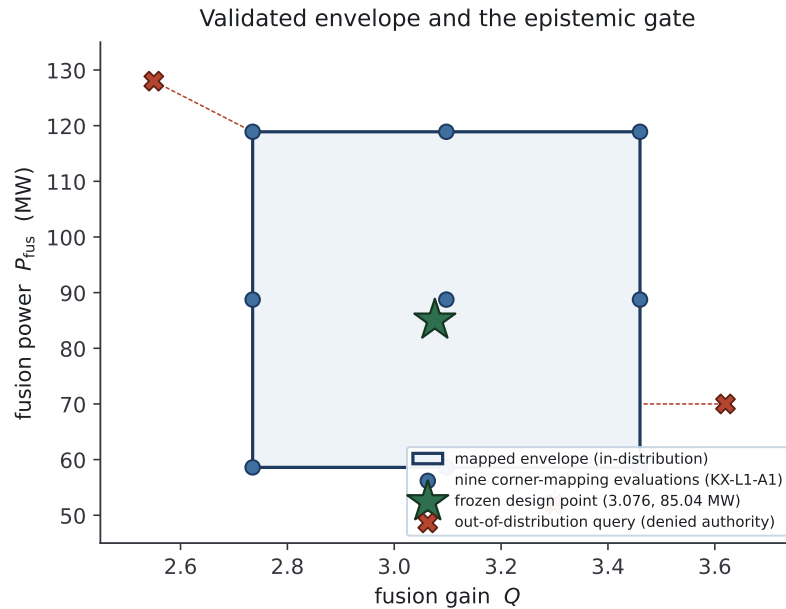


Figure 1. The validated envelope in the (Q, P_{fus}) plane. The bounding box of the mapped design-closure envelope (KX-L1-A1) is spanned by the nine corner-mapping evaluations; the frozen design point ($Q = 3.076$, $P_{\text{fus}} = 85.04$ MW) is interior. Out-of-distribution query points are flagged and denied model authority.

4.2 The reconstruction detector and graceful degradation

Figure 3 illustrates the reconstruction-detector principle of equations (7)–(8): the in-distribution reconstruction-error density is concentrated at low error, the held-out (out-of-distribution) regime

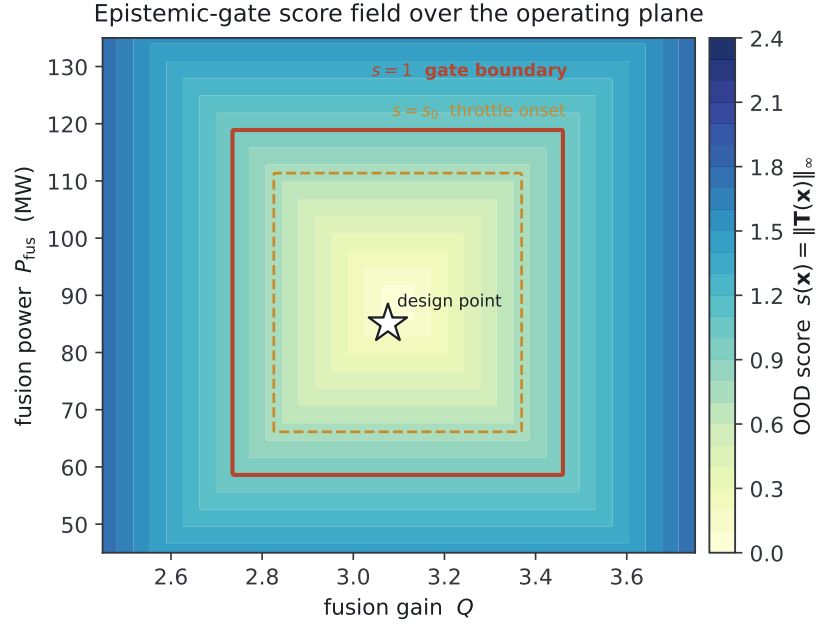


Figure 2. The epistemic-gate score field $s(\mathbf{x}) = \|\mathbf{T}(\mathbf{x})\|_\infty$ of equation (6) over the operating plane. The heavy contour $s = 1$ is the envelope boundary; the dashed contour $s = s_0$ is the throttle onset. The score is a defined gate function over the real mapped envelope, not measured data.

is displaced to high error, and the calibrated threshold τ at a 1% false-positive rate separates them, tripping the conservative fallback. Figure 4 shows the intended graceful-degradation behaviour of equation (9): detector discrimination declines slowly and monotonically as sensor channels are dropped, holding above a fallback-review floor over a wide range before the controller reverts.

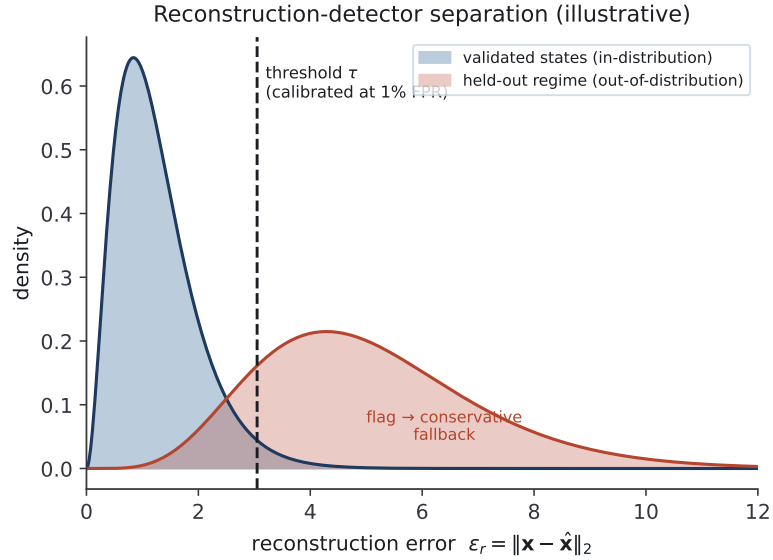


Figure 3. (Schematic.) Reconstruction-detector separation. In-distribution and held-out reconstruction-error densities with the calibrated threshold τ set at a 1% false-positive rate by equation (8). The distributions are illustrative of the detection principle; no measured values are plotted.

4.3 Calibrated uncertainty

Figure 5 shows the guarded confinement posterior of equation (14): $H_{98} = 1.007 \pm 0.030$, a tight Gaussian centred essentially on the calibration reference $H_{98} = 1.0$, with a 95% credible interval of $[0.948, 1.066]$. This is the calibrated width the authority gate A_σ reads through the epistemic term of equation (11).

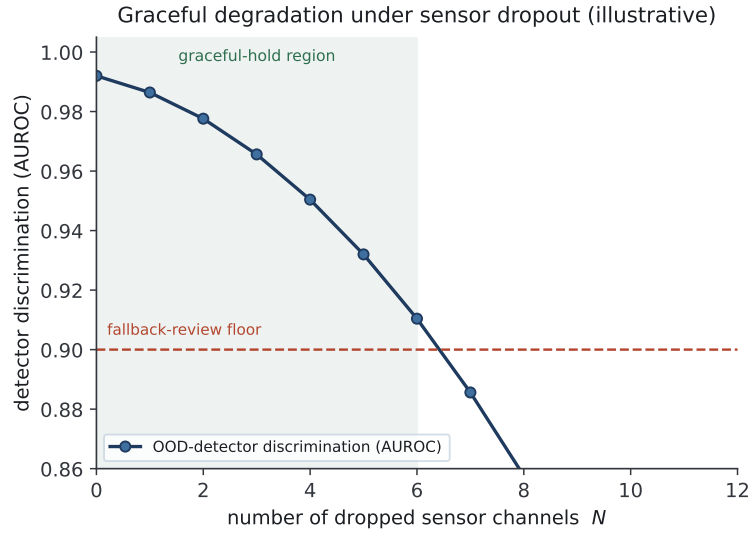


Figure 4. (Schematic.) Graceful degradation under sensor dropout. Detector discrimination versus the number of dropped channels N with the channel-indexed thresholds of equation (9); the shaded band marks the graceful-hold region above the fallback-review floor. Illustrative; no measured values are plotted.

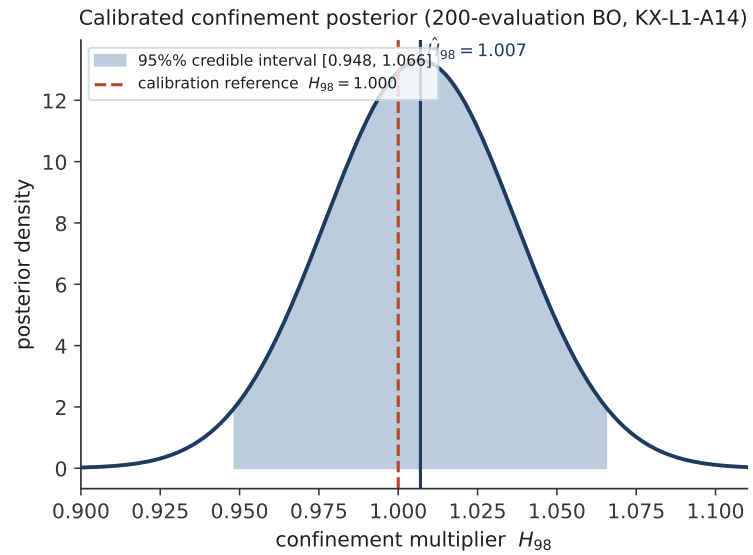


Figure 5. Calibrated confinement posterior $H_{98} = 1.007 \pm 0.030$ from the 200-evaluation surrogate-accelerated Bayesian-optimization campaign (KX-L1-A14). Shaded: the 95% credible interval; dashed: the calibration reference $H_{98} = 1.000$. Values are the frozen register anchor.

4.4 The composite authority gate

Figure 6 plots the gating law of §2.5: the piecewise authority-throttle $A_{\text{ood}}(s)$ of equation (15) (left), and the composite min-rule of equation (17) along a trajectory that leaves the envelope (right), where the OOD, ledger and uncertainty ceilings each cap authority independently and the minimum governs. Authority falls smoothly to the deterministic floor.

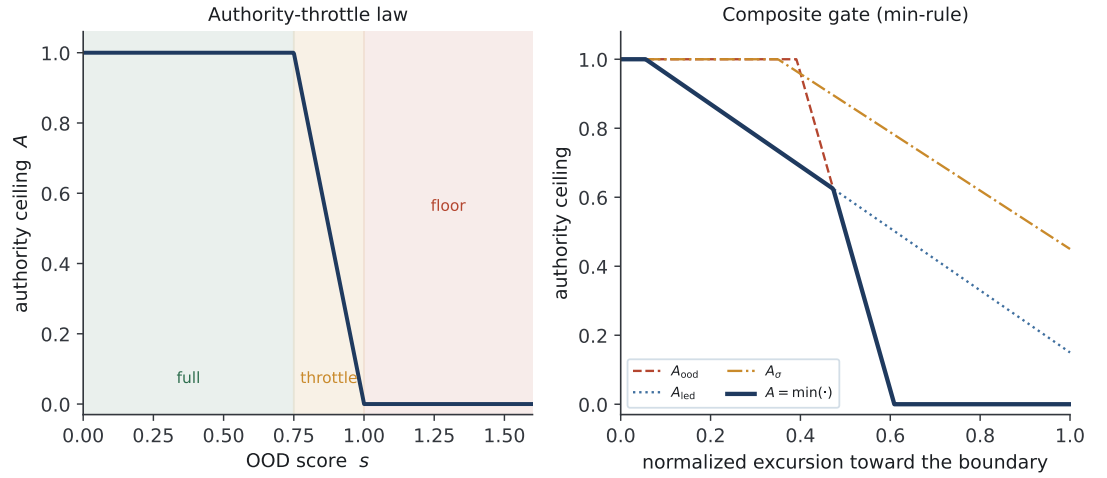


Figure 6. The authority-throttle law. *Left:* $A_{\text{ood}}(s)$ of equation (15) — full authority for $s \leq s_0$, a linear throttle to $s = 1$, and the deterministic floor beyond. *Right:* the composite $A = \min(A_{\text{ood}}, A_{\text{led}}, A_{\sigma})$ of equation (17) along an excursion toward the boundary; the most conservative gate wins. Both panels plot defined gate functions.

4.5 Discharged proofs and ledger support

Figure 7 records the formal verification status: the KX-L1-A6 suite discharges 19/19 anchors PASS across the MMS, GCI, Sobol and PCMM classes — the proofs the ledger monitor requires before certifying a lineage. Figure 8 shows the structure of the nine-device extrapolation-ledger support map of equation (10): each device contributes support across operating regions spanning the breeder and the burner, and the composite support density sets A_{led} point by point. Table 1 collects the epistemic-gating stack — each layer, its anchor, and the reproduced evidence behind it.

Formal V&V suite (KX-L1-A6): 19/19 anchors PASS



MMS: manufactured solutions · GCI: grid-convergence index · Sobol: variance-based sensitivity · PCMM: predictive-capability maturity

Figure 7. Formal verification-and-validation status (KX-L1-A6): 19/19 anchors PASS, grouped by the four verification classes of the suite (method of manufactured solutions; grid-convergence index; variance-based Sobol sensitivity; predictive-capability maturity model). The count is the frozen register anchor.

Table 1. The epistemic-gating stack — each layer, its frozen anchor or serial, and the reproduced evidence.

layer	anchor / serial	reproduced evidence
validated envelope	KX-L1-A1	Q 2.735–3.460, P_{fus} 58.6–118.9 MW, 9-pt grid
extrapolation ledger	KX-L1-A7	9-device post-diction, breeder + burner
calibrated uncertainty	KX-L1-A14	$H_{98} = 1.007 \pm 0.030$ posterior (200-eval BO)
discharged proofs	KX-L1-A6	formal V&V 19/19 anchors PASS
OOD detector	KFE-CF-SH-040	flags held-out regime; conservative fallback
ledger monitor	KFE-CF-SH-075	flags out-of-range operation; throttles authority

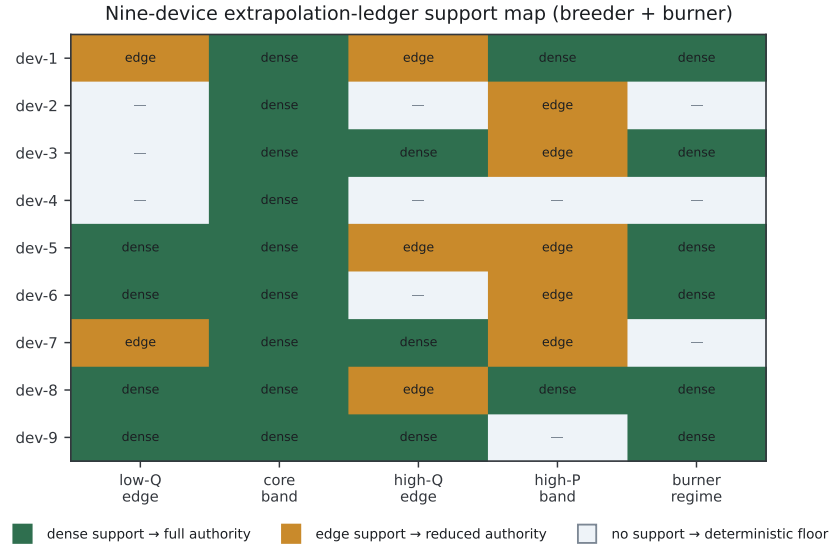


Figure 8. (Schematic.) The nine-device extrapolation-ledger support map (KX-L1-A7), spanning the breeder and the burner. Cells encode support class — dense (full authority), edge (reduced authority) or none (deterministic floor) — as computed from the kernel support of equation (10). The nine-device, breeder-plus-burner structure is grounded; the per-cell support pattern is illustrative.

4.6 Latency and the closed-loop rule

In closed loop the three layers compose into a single per-tick rule. The geometric OOD score (6) and the ledger lookup (10) are $O(nd)$ and $O(\sum_d |C_d|)$ evaluations that complete within one control tick on the twin node; the reconstruction detector (7) is a single GPU forward pass; the ensemble epistemic width (11) is read from the running forecast. Their minimum (17) sets the authority ceiling, and the blend (18) applies it. When the point is interior with dense ledger support and tight uncertainty, the learned controller acts at full authority; when any gate degrades, authority is throttled smoothly; when the point leaves the mapped envelope, the controller falls to the deterministic floor of equation (19). Authority is thereby bounded by evidence at all times.

5 Discussion

5.1 The gate against the OOD-detection literature

The two-detector design deliberately fuses complementary failure modes. A feature-space Mahalanobis score (5) is the strong baseline for detecting inputs far from the training manifold [9]; a reconstruction score (7) catches structured novelty that a low-dimensional distance can miss [16, 15]; and the geometric box score (6) ties both to the *physically* validated envelope rather than to an abstract density. Against the canonical baselines — softmax confidence [8], deep ensembles [10], MC-dropout [11] — the Kronos detector is distinguished less by a single AUROC number than by its coupling to a design-closure map with reproduced physics gates: the “in-distribution” set is not a data cloud but a verified operating envelope. This is the property that makes the detector’s verdict admissible in a nuclear safety case.

5.2 Calibration and the conformal guarantee

Modern networks are systematically over-confident [13], so a raw confidence is unsafe to gate on. The Kronos stack instead reads a *calibrated* width — the coverage condition of equation (12) holds the empirical coverage to 3% of nominal — and wraps it in the distribution-free conformal guarantee of equation (13), which holds in finite samples under exchangeability alone [18, 19]. The confinement posterior $H_{98} = 1.007 \pm 0.030$ is tight and centred, so the uncertainty gate A_σ is near unity at the design point and falls only as the epistemic term of equation (11) grows off-distribution. The separation of epistemic from aleatoric uncertainty [12, 17] is what lets the gate throttle on ignorance while tolerating irreducible noise. The abstention companion [28] develops the handing-back-to-a-floor decision in full.

5.3 Relation to learned fusion control and disruption prediction

Data-driven disruption predictors demonstrate that learned warnings can be accurate and transfer across machines [5, 6, 7], and reinforcement-learning controllers now shape and stabilise real

plasmas [1, 2]. Those results make the *capability* case; the epistemic-gating layer makes the *admissibility* case that must sit beneath them, and it is the natural home for the cross-machine transfer uncertainty a disruption model inherits when it is deployed off its training devices. The bound-limited Bayesian-optimization excursion of §2.4 ($Q \approx 8.3$ under widened priors) is a concrete instance of the confident off-distribution extrapolation the gate refuses: the optimizer is free to *propose* it, but the composite ceiling of equation (17) denies it authority because it lies outside the mapped envelope, the ledger has no support there, and the epistemic width has blown up. The maturity-gated actuation-authority argument [30] binds this runtime ceiling to the verification credibility of §3, and the global sensitivity-and-UQ survey of the frozen anchors [34] quantifies how the envelope’s error bar propagates.

5.4 Why the min-rule

The composite (17) is a minimum, not a product or an average, by design: authority should be capped by the *worst* evidence gate, so that a single failing channel — an unfamiliar regime, thin ledger support, or a blown-up epistemic width — is sufficient to throttle, and no gate can be “bought back” by another. This is the conservative choice a nuclear safety case requires, and it makes the gate’s behaviour monotone and auditable: reducing any evidence source can only lower authority, never raise it.

6 Limitations

One honest gate remains. The reconstruction detector’s threshold family $\{\tau_O\}$ of equations (8)–(9) is calibrated against the twin’s validated state distribution; tightening it against *machine* diagnostics is the one confirming step, and it is the natural first use of live data as the machine accrues operating hours. Until then the gate is conservative by construction: an unfamiliar state defaults to the floor, so the calibration gap can only make the controller more cautious, never less safe. The ledger support kernel (10) is likewise resolved from the nine-device post-diction base and will sharpen as operating cases accumulate.

7 Conclusion

The Kronos autonomous controller carries an explicit epistemic-gating layer, stated here as mathematics and grounded in reproduced evidence. The validated envelope is *mapped* over a nine-point design-closure grid (Q 2.735–3.460, P_{fus} 58.6–118.9 MW) with the frozen design point (3.076, 85.04 MW) interior; two complementary detectors — a reconstruction/feature OOD score (SH-040) and a nine-device extrapolation ledger spanning the breeder and burner (KX-L1-A7; monitor SH-075) — measure how far the live point has moved outside it; the guarded confinement model carries a calibrated posterior $H_{98} = 1.007 \pm 0.030$ with a distribution-free conformal coverage guarantee and passes 19/19 formal verification anchors; and a single composite min-rule throttles the learned controller smoothly to a certified control-barrier floor as the point approaches, and then leaves, the envelope. This is the credibility layer beneath the autonomy — authority that knows, and respects, the limit of its own evidence — and it converts the safety case from “trust the model everywhere” into “bound the model by what has been verified.”

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Reproducibility. The gating anchors are frozen entries in the Kronos Physics De-Risking Register [27] (doi:10.5281/zenodo.22133057): the validated envelope (KX-L1-A1); the nine-device extrapolation ledger (KX-L1-A7); the calibrated confinement posterior $H_{98} = 1.007 \pm 0.030$ (KX-L1-A14); and the formal verification suite 19/19 (KX-L1-A6). Each is regenerable from the archived closure, UQ and verification runs, and every figure is produced by the deposited generation script `make_ood_ledger_figs_2p79.py` from the frozen anchor values. This paper is archived at doi:10.5281/zenodo.22645819 and its official version is at <https://www.kronosfusionenergy.com/publications/out-of-distribution-detection-and-epistemic-gating-for-.html>.

Data availability

The envelope grid, the ledger schema, the detector calibration inputs and the confinement posterior used for figures 1–8 are archived with the deposit at doi:10.5281/zenodo.22645819, which references the Kronos 2026 series — including the calibrated-uncertainty and abstention result [28], the runtime-assurance provenance ledger [29] and the AI-and-quantum-control overview [36]. All quantitative values trace to the register [27] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work (KRONOS-CTRL) have been filed.

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